

Corporate Finance Ph.D. Seminar Final Exam 2024

Question 1 (Institutional Investors)

Crane, Koch, and Michenaud (2019) define cliques as groups of institutional investors connected through overlapping large ownership stakes (greater than 5%) in at least one firm, suggesting these groups coordinate activities such as voting or influencing firm policies. Cross investments could arise from portfolio size, indexing behavior, or perhaps personal links. The latter is the most interesting. In order to isolate the effect of personal connections to cross investment, proxies for personal connections must be identified and measured. Cohen et al. (2008) show that mutual fund managers with the same alma mater are more likely to hold similar stocks. This suggests that social ties influence investment decisions. Hong et al. (2005) find that mutual fund managers in the same city exhibit correlated portfolio choices, potentially due to informal interactions. Other interesting proxies include shared board memberships and shared employment history. Shared board memberships could facilitate direct manager interactions, as explored in studies of interlocking directorates (Davis, 1991). Although it is less commonly applied to institutional investors. Managers who previously worked at the same firm may maintain professional ties, influencing coordinated investments (Hochberg et al., 2007).

To isolate the effect of personal links, shared board memberships and geographic proximity are practical proxies with available data. Portfolio size and indexing factors must be controlled. For portfolio size, assets under management or the number of stocks may be suitable controls. Larger portfolios will increase the probability of a random overlap. (Crane et al., 2019). Indexing is identified by whether an institution is an index fund or by the correlation of its portfolio with a market index. Crane et al. (2019) suggest excluding index funds or controlling for indexing fraction to isolate active coordination.

Based on this framework, two hypotheses are proposed. Hypothesis 1: *Personal links among institutional investors increase the likelihood of their institutions forming cliques, beyond what is explained by portfolio size or indexing.* The rationale here is that personal connections facilitate coordination, leading to deliberate cross-investments. Hypothesis 2: *The effect of clique ownership on governance outcomes is stronger when cliques have denser personal links among members.*

The sample for this study will be US firms and institutional investors from 1980 to 2013, to be consistent with Crane et al. (2019). The data will come from Thomson Reuters 13F for

holdings data. The shared board membership will come from BoardEx, linking institutional managers to boards of portfolio firms. The geographic proximity data will come from the headquarters location from 13F filings or institutional records. The control variables will be Aum and number of stocks held (size control), index fund status or portfolio-index correlation (indexing control), and investment style similarity (measured by sector weights).

The outcome variables will be clique membership (measured using Crane et al.'s Louvain algorithm), Number of overlapping greater than 5% stakes, and governance outcomes (Voting data from ISS).

To test Hypothesis 1, our data will be on an institution-pair-year. For example, institutions i and j in year t . The first dependent variable will be $\text{SameClique}_{\{i,j,t\}}$ which equals a 1 if institutions i and j are in the same clique in year t , and a zero otherwise. The next dependent variable will be $\text{NumOverlap}_{\{i,j,t\}}$. This variable is the number of firms where both hold greater than 5% stakes in year t . The main independent variables will be $\text{SharedBoard}_{\{i,j,t-1\}}$ (which will be a 1 if managers from i and j served on the same board in $t-1$) and $\text{GeoProx}_{\{i,j,t-1\}}$ (inverse distance between headquarters). Here is the model specification:

$$\text{NumOverlap}_{i,j,t} = \alpha + \beta \cdot \text{SharedBoard}_{i,j,t-1} + \gamma \cdot \ln(\text{NumLargeStakes}_i \cdot \text{NumLargeStakes}_j) + \delta \cdot \text{Controls} + \epsilon_{i,j,t}$$

The controls will be AUM of i and j , index fund indicators, and portfolio similarity. The number of large stakes variable is a gravity model that accounts for expected overlaps based on size. There will be year fixed effects and standard errors will be clustered by institution pair.

To test for hypothesis two, the data will be on a firm-year level (for example, firm k in year t). The dependent variable will be $\text{VotesAgainst}_{\{k,t\}}$. This is the fraction of votes against bad proposals. $\text{CliqueOwnership}_{\{k,t\}}$ is the first independent variable, and measures the total clique ownership in firm k (from Crane et al., 2019). $\text{CliqueLinkDensity}_{\{k,t\}}$ is the other independent variable, and is the fraction of institution pairs in cliques owning k with shared board memberships. Firm size, institutional ownership, and blockholder presence will be the control variables along with firm and year fixed effects. Here is the model specification:

$$\text{VotesAgainst}_{k,t} = \alpha + \beta_1 \cdot \text{CliqueOwnership}_{k,t} + \beta_2 \cdot (\text{CliqueOwnership}_{k,t} \cdot \text{CliqueLinkDensity}_{k,t}) + \gamma \cdot \text{Controls} + \epsilon_{k,t}$$

Here, a positive β_2 indicates stronger governance effects when cliques have denser personal links.

Causality and identification concerns are addressed by using lagged personal link variables. This mitigates the possibility that clique formation drives board membership. For Hypothesis 1, the instrument variable SharedBoard is a strong instrument, assuming it affects overlaps only through connections.

Question 2)

Climate change and the increasing strictness of environmental regulations have become central features of modern business. In the United States and Europe, heightened awareness and regulations have reshaped how firms approach strategy, operations, and risk management. In the context of M&A, these developments have significant implications. For example, changing risk profiles and valuation. Firms with high carbon-producing activities or other environmental effects may face increased regulatory scrutiny, leading to higher compliance costs and potential penalties. Such risks are increasingly factored into valuations during the M&A process. Acquirers now account for not only traditional financial metrics but environmental factors that could affect the future performance of the firm. Another implication of climate change on M&A is how it affects the due diligence process. Investors are scrutinizing environmental performance indicators (ESG, historical environmental violations, sustainability practices, etc.). These indicators can influence the perceived synergies in post-merger integration. Investors are becoming more conscious of climate risks, and market expectations are shifting. Deals that signal strong environmental stewardship may attract higher premiums, while those associated with environmental risk might command discounts. This dynamic changes the bidding strategies during deal formation and anticipated post-merger performance. The intersection of environmental concerns and corporate decisions has gathered significant attention. Porter and van der Linde (1995) argue that well-designed environmental regulations can stimulate innovation and improve competitiveness. This is now known as the Porter Hypothesis. Their work suggests that stringent regulations can drive firms to innovate, and might not always pose a cost burden. However these benefits can be unevenly distributed. Hall et al. (2023) develop a firm-level metric of supply-chain climate exposure that predicts stock returns and captures firms' sensitivity to climate shocks. In the context of M&A, targets with high Hall-exposure should command larger discounts, as acquirers anticipate post-deal cash-flow volatility and higher hedging costs. Sautner et al. (2023) document a 3.7 percent annualized premium for firms with high idiosyncratic climate risk. This implies that acquirers face a measurable cost of carrying "brown" assets. Pankratz & Schiller (2024) show that suppliers' physical-risk exposure (e.g. heat stress

days) reduces customer operating income and forces buffer-inventory strategies. In an M&A context, targets with Pankratz-measured exposure may underperform post-merger.

To test these effects, I propose hypothesis 1: Combined CARs decrease with the net supply chain exposure of bidder + target. This is inspired by Hall et al. (2023). I also propose hypothesis 2: Acquirers buying targets with high idiosyncratic climate risk earn negative CARs. This is derived from Sautner et al. (2023). The last hypothesis is : Targets with higher Pankratz-measured supplier physical-risk exposure exhibit lower post-deal operating income growth and BHARs. This hypothesis is inspired by Pankratz & Schiller (2024). The sample for this study will include U.S. domestic public-firm mergers from 2005 to 2025 with a deal value greater than 50 million. This should give me a large enough sample to derive the results from. The key variables will be HallExposure, SautnerRisk and PhysRisk. HallExposure is the supply chain climate exposure as measured by Hall et al. (2023). It includes scope 1 + 2 and downstream supply chain exposures. SautnerRisk is idiosyncratic climate-risk score inspired by Sautner et al. (2023). This can be created with either the exact model with the exact weights that they used, or my own replication of the model. PhysRisk is the Pankratz-Schiller supplier exposure. This is calculated by geocoding suppliers and assembling climate hazard data such as heat stress days, heavy precipitation events, and drought indices. The exact model will be as follows:

$$\underbrace{\text{DealSuccess}_i}_{\text{CAR}_i(-2,+2) \text{ or } \text{BHAR}_i(T)} = \alpha + \beta_k \text{ClimateMetric}_i^{(k)} + \gamma' \mathbf{X}_i + \varepsilon_i$$

Where k is represents the different climate metric index. Climate metric 1 is HallExposure, climate metric 2 is SautnerRisk, and climate metric 3 is PhysRisk. Xi are the deal controls such as deal value, cash-vs-stock dummy, relatedness (via NAICS-4 distance), hostile / friendly indicator, etc. I will apply year and industry fixed effects. Acquirer fundamentals will be used as well such as leverage ratio, Tobin's q, ROA, and market-to-book ratio. I will use CAR(-2,+2) as the measure of deal success for hypothesis 1 and 2. I expect HallExposure and SautnerRisk to have coefficients less than 0. This would indicate that these climate metrics adversely affect deal success. To test PhysRisk, I need a longer timeline. I will instead use BHAR to test for the long term buy and hold return of this deal. I expect the coefficients of PhysRisk to be less than 0 also.

The SEC disclosure rule is a great quasi-experiment. I will introduce a post-sec dummy for deals announced on or after March 7, 2024. Interacting this with HallExposure and SautnerRisk will test whether mandatory reporting led to a sharper market reaction to carbon and idiosyncratic risk. Robustness checks will include a propensity score matched dataset, instrument variables (using historical regional temperature variability as an instrument for HallExposure), and shorter / longer CAR windows.

The deal data will come from SDC Platinum. Return and financial data will come from CRSP and Compustat. Supply chain links will come from FactSet Revere, and climate hazards will come from the NOAA.

By combining quasi-experimental variation from the SEC's 2024 disclosure rule, matched-control and IV approaches, and robustness checks, this design will deliver credible identification of how transition and idiosyncratic climate risks drive merger outcomes. The results will inform dealmakers and policymakers on the valuation importance of climate metrics and the efficacy of mandatory disclosure.

Question 3

Bharath & Hertz (2019) show that in the United States, firms tend to substitute toward relationship-bank debt when the outside equity governance channel is weak. I will now show how I would test whether this substitution is stronger, weaker, or potentially reversed for non-US firms.

External governance encompasses everything outside the internal boardrooms that disciplines managers. For example, takeover market pressure, activist shareholders, analyst and media scrutiny, and the country's securities laws. There are several channels that could explain the trade-off changing outside the U.S.

The first is that minority investor protection is weak in many emerging market countries. Therefore, equity governance is weaker and cannot bite. In these situations, firms should rely more on monitoring by relationship banks. This would indicate a stronger substitution effect. This is supported by Djankov et al. (2008) who show that the anti-self-dealing index shows big cross-country gaps.

Another channel is the heterogeneous nature of creditor rights. Where foreclosure and collection are costly, banks cannot monitor effectively. Debt cannot substitute. This leads to a weaker substitution effect.

Ownership concentration is higher abroad. Controlling owners block hostile takeovers but often cultivate house banks that sit on the board. In this setting, bank debt can become a complement to concentrated ownership rather than a substitute.

With all of these channels in mind, I propose three hypotheses. H1 is regarding the investor-protection channel. In countries with low anti-self-dealing scores, the negative relation between external governance strength and the share of bank debt in total debt is more negative than in high-protection countries. This will be tested with the following regression.

$$\text{BankDebtShare}_{ict} = \alpha + \beta_1 \text{ExtGov}_{ict} + \beta_2 (\text{ExtGov}_{ict} \times \text{LowASD}_c) + \gamma' X_{ict} + \mu_i + \lambda_t + \varepsilon_{ict}$$

This regression shows the effect of external governance (as measured by institutional ownership in this study) and low anti-self-dealing scores on bank debt share. The subscripts are i = firm, c = country, and t = year. Controls are $X(ict)$ which are to be log assets, tangibility, profitability, market-to-book, and industry dummies. This regression includes firm and year fixed effects as well. The standard errors are clustered at the firm level. The coefficient of interest here is β_2 . I expect β_2 to be less than zero. This would mean that where minority-investor protection is weak, the negative slope between external governance and bank-debt reliance becomes more negative, indicating stronger substitutability.

Hypothesis two is regarding the creditor rights channel. The idea is that the substitutability documented in the United States weakens or disappears in countries with below-median creditor-rights indices. This hypothesis will be tested with the following regression:

$$\text{BankDebtShare}_{ict} = \alpha + \beta_1 \text{ExtGov}_{ict} + \beta_3 (\text{ExtGov}_{ict} \times \text{LowCredRights}_c) + \gamma' X_{ict} + \mu_i + \lambda_t + \varepsilon_{ict}$$

Here, ExtGov is measured as institutional ownership as well. LowCredRights comes from the Creditor-Rights index from Djankov et al. (2007) paper. The dummy is constructed as a 1 if the Creditor Rights Index is less than or equal to the median across all sample countries, and a 0 otherwise. The same controls from H1 are used. Here, we expect β_3 to be positive. This would indicate that where creditor rights are weak, banks cannot monitor effectively. The negative U.S. slope is muted or offset. This would indicate weaker substitutability.

Hypothesis 3 tests the Ownership-structure interaction. The hypothesis is that among firms with a controlling shareholder (more than 30 percent voting power), bank debt complements rather than substitutes for external governance.

$$\text{BankDebtShare}_{ict} = \alpha + \beta_1 \text{ExtGov}_{ict} + \beta_4 (\text{ExtGov}_{ict} \times \text{ControlOwn}_{ict}) + \gamma' X_{ict} + \mu_i + \lambda_t + \varepsilon_{ict}.$$

Here, ExtGov is the same measure of institutional ownership, and ControlOwn is a 1 if there is a controlling shareholder. The same controls from H2 and H1 are used here. The sign we expect is for B4 to be greater than 0. The intuition behind this is that for firms with a controlling shareholder, relationship banks often partner with insiders and bank debt shifts from being a substitute to a complement. So the interaction flips or attenuates the baseline negative slope.

The sample will include public firms in 40+ countries from around the world. This will come from compustat global. Bank-debt share will be calculated as loans divided by total debt. This will come from DealScan. The identification strategy will use firm and year fixed effects to absorb time-invariant heterogeneity and macro trends. For an instrumental variable, I will interact global bank-capital stocks with pre-crisis firm relationships to instrument bank-debt supply. (Hasan et al. 2017 uses a similar trick).

For further robustness, propensity score matched firms will be used to match firms based on size, industry, profitability. I will also replace bank-debt share with loan-spread or covenant-intensity measures.

I expect this study to demonstrate that substitution intensifies when minority protections are weak, vanishes when creditor rights are weak, and flips under concentrated ownership. This would extend Bharath and Hertzels U.S. findings to a global context and highlight how legal origin, creditor enforcement, and ownership structure jointly shape firm's financing choices.